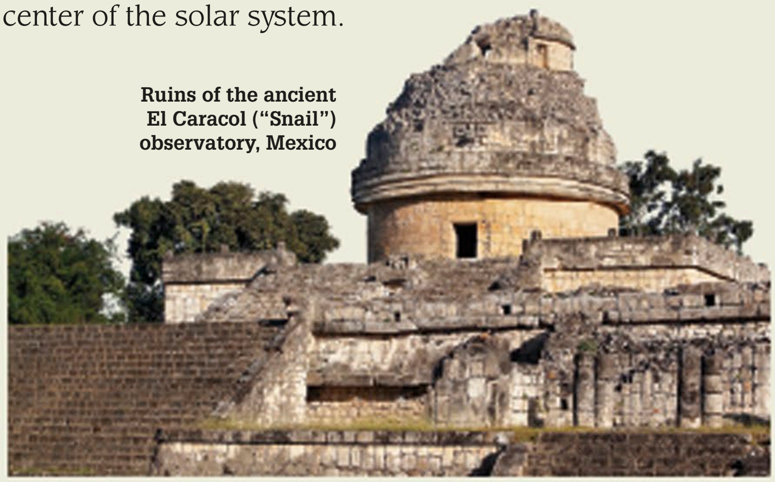
Astronomy

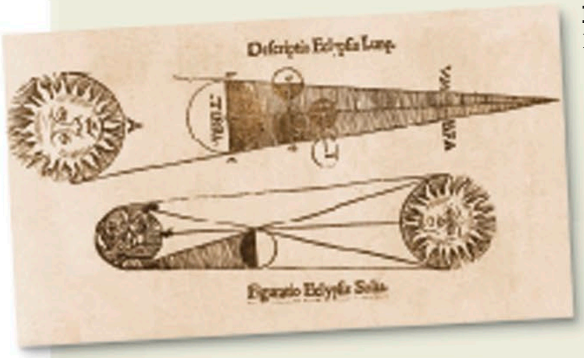
Many ancient peoples, such as the Maya of Central America, the Chinese, Indians, and Babylonians, tried to make sense of the motions of stars and planets. From the 4th century BCE, the Greeks developed models to explain why planets changed position in the sky. Not until the 16th century did astronomers realize that the Sun, not Earth, is the center of the solar system.



Ruins of the ancient El Caracol ("Snail") observatory, Mexico

Ancient observatories

The Maya, whose culture was at its peak from 250–900 CE, built observatories, such as El Caracol ("Snail," named for its shape) at Chichén Itzá, in Mexico. They accurately calculated the length of the year and recorded the movements of the planet Venus.



Eclipses

In the 13th century, English monk Johannes of Sacrobosco reproduced calculations made by the astronomer Ptolemy of Alexandria in the 2nd century CE. These showed how the movement of the Moon in front of the Sun causes an eclipse.

Solar and lunar eclipses, from Sacrobosco's book *De sphaera mundi* (Sphere of the World)

Astronomers using astrolabes, in the *The Travels of Sir John Mandeville*, 1356

Key events

C 500 BCE

Babylonian astronomers created the zodiac, dividing the sky into 12 equal zones through which the Sun and the planets appeared to travel.

C 350 BCE

Eudoxus of Cnidus, a Greek mathematician, devised the first model of the solar system based on concentric spheres. He used 27 spheres, several for each planet, to explain irregular orbits.

C 280 BCE

Aristarchus of Samos suggested that Earth orbits around the Sun. Fellow Greek astronomers and scholars criticized his ideas, which were not accepted.

C 240 BCE

Eratosthenes of Cyrene (now in Libya) accurately measured the circumference of Earth by comparing shadows cast by the Sun in two different locations.